Laboration 1: Two Competing Species - Coupled Logistic Maps

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1. Problem statement [1]

In this lab we will be investigating a coupled logistic map. Logistic maps are an example for a dynamical system, a system that is governed by a rule that determines how the system evolves from one time step to the next. Additionally to that rule, the system needs a state space and a notion of time. As we are working with discrete time scales, the next step is obtained as follows: $x_{n+1} = f(x_n)$.

The goal of the lab is to observe how this system behaves for different initial conditions. There are many different features we can use to study its behaviour. First, there are the fixed points. These are the points where the update to the next state gives the same state again: $f(x^*) = x^*$. Next, a system can have periodic points. In this case the system returns to the same state after a certain amount of iterations ($f^p(x) = x$ for $p \geq 2$). The stability of a fixed point can also be of interest. To investigate this, we can look at what happens when a small deviation δ from our fixed point occurs:

$$\delta = x_n - x^*$$

Using a Taylor expansion we can write the next time step as

$$x_{n+1} - x^* \approx f(x^* + \delta) - f(x^*) + f'(x^* + \delta) * \delta$$

which we can simplify to be

$$x_{n+1} - x^* \approx f'(x^*)(x_n - x^*)$$

From this expression we can deduce whether a deviation from the fixed point will become larger or smaller.

If $|f'(x^*)| < 1$ the deviation is getting smaller, the fixed point is stable.

If $|f'(x^*)| > 1$ the deviation is getting larger, the fixed point is unstable.

Now, we would like to introduce the tasks of lab 1. The discrete-time model of two coupled logistical maps we are studying is supposed to simulate two competing populations. The size of each population is, in comparison to uncoupled logistic maps, not only determined by its own size in the preceding generation but also by the size of the competing population in the preceding generation through the coupling term.

In the first task we will investigate the fixed points. We will look at several cases: the trivial fixed point, the diagonal case where $x^* = y^*$ and the asymmetric case where the growth parameters of both populations are unequal. Next, we will investigate if any periodic points with period 2 or 3 exist. This will be done for several parameter sets as the behaviour of the model changes for different growth parameters. To visualise the long-time orbit with different growth parameters we will make phase portraits. Lastly, to visualise the different regimes we will plot the bifurcation cascade. This shows the x-y values that a certain starting point approaches for different growth parameters.

All of these tasks together will give us an overview of the behaviour of the system: do we have fixed points and/ or periodic points? Are they stable? How does the behaviour differ for different sets of parameters?

2. Theoretical Background

In this section we present the theory needed to solve the tasks of the lab.

2.1. Problem 1 and 2: Fixed Points, Periodic Points, and Newton's Method

A discrete-time dynamical system describes the evolution of a state variable through successive iterations of a given map. In this project's model, the state of the system n is described by the vector

$$z_n = (x_n, y_n),$$

where x_n and y_n denote the normalized densities of two interacting populations. The evolution of the system is determined by a map F, such that

$$z_{n+1} = F(z_n).$$

2.1.1. Fixed Points

A fixed point of a map is a state that remains unchanged under one application of the map. Thus, a point (x^*, y^*) is a fixed point of F if

$$F(x^*, y^*) = (x^*, y^*).$$

In this context, fixed points correspond to equilibrium states, where the reproductive dynamics and the interaction between the two populations are balanced. This is reflected in a constant population density from one generation to the next. So, by definition, in order to get the fixed points, one has to find the roots of the system.

2.1.2. Periodic Points

Periodic points generalize the concept of fixed points. A point (x, y) is said to be periodic with period p if it returns to itself after p applications of the map:

$$F^p(x, y) = (x, y),$$

For example,

$$F^2(x, y) = F(F(x, y)),$$

and

$$F^3(x, y) = F(F(F(x, y))).$$

A period-2 and 3 point therefore satisfies, respectively,

$$F^2(x, y) = (x, y)$$

or

$$F^3(x, y) = (x, y),$$

meaning that the orbit repeats after two respectively three iterations.

It is important, however, to distinguish between a point satisfying $F^p(x, y) = (x, y)$ and a point having *minimal period p*. The minimal period is the smallest positive integer k such that

$$F^k(x, y) = (x, y).$$

For example, every fixed point satisfies

$$F^2(x, y) = (x, y),$$

but it should not be classified as a genuine period-2 point, because it already returns to itself after one iteration. Therefore, after solving the equation $F^p(x, y) = (x, y)$, it is necessary to verify that the solution does not have a smaller period.

2.1.3. Reformulation as a Root-Finding Problem

The numerical computation of periodic points can be formulated as a nonlinear root-finding problem. This problem is solved with the aid of Newton's method and, for that, it is necessary to adapt the problem. For a given period p, the function is defined as

$$G_p(x, y) = F^p(x, y) - (x, y).$$

To determine if a point is periodic, the following condition has to be met:

$$G_p(x, y) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This way, the problem is reduced strictly to finding the zeros of the function without the occurrence of repeated orbits. For $p = 2$, it solves

$$G_2(x, y) = F^2(x, y) - (x, y) = 0,$$

while for $p = 3$, it solves

$$G_3(x, y) = F^3(x, y) - (x, y) = 0.$$

This reformulation is useful and necessary because the iterated maps F^2 and F^3 generate nonlinear systems that are generally too complicated to solve analytically. A numerical method is therefore required.

2.2. Problem 3: Phase Portraits

In general, a phase portrait displays the evolution of the state of a dynamical system by plotting subsequent states in the same diagram. For one-dimensional continuous systems the axes would usually be $\dot{x}$ and x. For more than one dimension or state variable one has to choose them to be well representable in two dimensions. In the system examined here we have an iterated map with two dimensions which are also the axes in the phase portrait plot.

If an attractive fixed point exists in the system, later iterations would be ever closer to that fixed point than previous iterations. A stable orbit of length n could not be distinguished from n fixed points being present and would be needed to be further examined by the other methods employed in this report. Chaotic conditions can be inferred when there is no discernible pattern in the phase portrait. Pattern could here refer to either convergence or regularity.

3. Model and parameters

The model under investigation is

$$x_{n+1} = (1-\varepsilon)r_1 x_n(1-x_n) + \varepsilon r_2 y_n(1-y_n) \tag{3.1}$$
$$y_{n+1} = (1-\varepsilon)r_2 y_n(1-y_n) + \varepsilon r_1 x_n(1-x_n) \tag{3.2}$$

which is equivalent to a non-linear perturbation of a linear map:

$$\mathbf{x}_{n+1} = \mathbf{F}(\mathbf{x}_n) = A\mathbf{x}_n + \mathbf{f}(\mathbf{x}_n) \tag{3.3}$$

with

$$A = \begin{bmatrix} (1-\varepsilon)r_1 & \varepsilon r_2 \\ \varepsilon r_1 & (1-\varepsilon)r_2 \end{bmatrix}, \quad \mathbf{f}(\mathbf{x}) = A\mathbf{g}(\mathbf{x}), \quad \mathbf{g}(\mathbf{x}) = -x^2\mathbf{e}_x - y^2\mathbf{e}_y \quad \mathbf{x} = \mathbf{x}_n \tag{3.4}$$

so

$$\mathbf{F}(\mathbf{x}) = A\mathbf{x} + \mathbf{f}(\mathbf{x}) = \begin{bmatrix} (1-\varepsilon)r_1 & \varepsilon r_2 \\ \varepsilon r_1 & (1-\varepsilon)r_2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} (1-\varepsilon)r_1 & \varepsilon r_2 \\ \varepsilon r_1 & (1-\varepsilon)r_2 \end{bmatrix}\begin{bmatrix} -x^2 \\ -y^2 \end{bmatrix}. \tag{3.5}$$

Looking for fixed points amounts to the root finding problem

$$\begin{aligned} \mathbf{G}(\mathbf{x}) &= \mathbf{F}(\mathbf{x}) - \mathbf{x} \\ &= A\mathbf{x} + \mathbf{f}(\mathbf{x}) - \mathbf{x} \\ &= A\mathbf{x} + A\mathbf{g}(\mathbf{x}) - \mathbf{x} \\ &= A(\mathbf{x} + \mathbf{g}(\mathbf{x})) - \mathbf{x} \\ &= 0 \end{aligned} \tag{3.6}$$

$$\Rightarrow \mathbf{x} + \mathbf{g}(\mathbf{x}) = A^{-1}\mathbf{x} \tag{3.7}$$

$$\iff \mathbf{g}(\mathbf{x}) = (A^{-1} - I)\mathbf{x} \tag{3.8}$$

$$A^{-1} = \frac{1}{\det(A)}\operatorname{adj}(A) = \frac{1}{\det(A)}\begin{bmatrix} (1-\varepsilon)r_2 & -\varepsilon r_1 \\ -\varepsilon r_2 & (1-\varepsilon)r_1 \end{bmatrix}^T \tag{3.9}$$

$$\det(A) = (1-\varepsilon)^2 r_1 r_2 - \varepsilon^2 r_1 r_2 = (1-2\varepsilon)r_1 r_2 \tag{3.10}$$

$$\iff \begin{bmatrix} -x^2 \\ -y^2 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} \tag{3.11}$$

$$\begin{cases} a = \frac{1-\varepsilon}{r_1(1-2\varepsilon)} - 1 = \frac{\varepsilon-1}{2r_1(\varepsilon-1/2)} - 1 \\ b = \frac{-\varepsilon}{r_1(1-2\varepsilon)} = \frac{\varepsilon}{2r_1(\varepsilon-1/2)} \\ c = \frac{-\varepsilon}{r_2(1-2\varepsilon)} = \frac{\varepsilon}{2r_2(\varepsilon-1/2)} \\ d = \frac{1-\varepsilon}{r_2(1-2\varepsilon)} - 1 = \frac{\varepsilon-1}{2r_2(\varepsilon-1/2)} - 1 \end{cases} \tag{3.12}$$

$$\begin{cases} x^2 + ax + by = x(x+a) + by = 0 \\ y^2 + cx + dy = cx + y(y+d) = 0 \end{cases} \tag{3.13}$$

Thus the fixed points belong to the intersections of 2 parabolic surfaces. The simple nullcline (that is if we ought to find a fixed point it is necessary for it to lie on this curve) is given by the first equation above

$$y(x) = \frac{-x^2 - ax}{b} \tag{3.14}$$

Inserting Eq (3.14) in the lower equation in (3.13) gives the following quartic polynomial:

$$p(x) = x^4 + 2ax^3 + (a^2 - bd)x^2 + b(cd - ad)x \tag{3.15}$$

From this we can readily factor x and thus $\mathbf{x}_1 = (0, 0)$ is the trivial root. This leaves us with

$$q(x) = x^3 + 2ax^2 + (a^2 - bd)x + bd(c - a) \tag{3.16}$$

and the condition $q(x) = 0$ as the relevant requirement for a fixed point. Since $\deg(q) = 3$ and $y(x)$ is a one-to-one mapping for $b \neq 0$, we can have, at most, four fixed points. In our case, we might have even less than four fixed points since we also treat population rates on the square $[0, 1]^2$.

3.1. Population equilibrium & Fixed points

3.1.1. Symmetric fixed points

If we consider the line parameterization $x = y = t$ where $t \in [0, 1]$, which is the linear subspace of population rates of equal magnitude, we could ask if there is a rate t^* such that $\mathbf{G}(t^*) = 0$? Let $x = y = t$, then

$$\begin{aligned}
\mathbf{G}(\mathbf{x}) &= A\mathbf{x} + A\mathbf{g}(\mathbf{x}) - \mathbf{x} = 0 && (3.17)\\
&= A\mathbf{x} + A\mathbf{g}(\mathbf{x}) - \mathbf{x} && (3.18)\\
&= \mathbf{x} + \mathbf{g}(\mathbf{x}) - A^{-1}\mathbf{x} && (3.19)\\
&= \mathbf{g}(\mathbf{x}) + \mathbf{x} - A^{-1}\mathbf{x} && (3.20)\\
&= -t^2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{1}{(1 - 2\varepsilon)r_1 r_2} \begin{bmatrix} (1-\varepsilon)r_2 & -\varepsilon r_2 \\ -\varepsilon r_1 & (1-\varepsilon)r_1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} && (3.21)\\
&= -t^2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + t \begin{bmatrix} 1 - \frac{1}{r_1} \\ 1 - \frac{1}{r_2} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. && (3.22)
\end{aligned}$$

If we disregard the trivial fixed point $t \neq 0$, then we get that

$$\begin{cases} t = 1 - \frac{1}{r_1} \\ t = 1 - \frac{1}{r_2} \end{cases} \tag{3.23}$$

which can hold simultaneously only if $r_1 = r_2 = r$. Thus, except for the trivial fixed point, we can have a fixed point of equally large population rates if the growth factors of both populations are equal. The second fixed point in the diagonal case is thus along a line through the origin $\mathbf{x} = \lambda(r)\mathbf{v}$:

$$\mathbf{x}_1(r) = \left(1 - \frac{1}{r}\right) \begin{bmatrix} 1 \\ 1 \end{bmatrix} \tag{3.24}$$

with the constraint $1 \leq r$ since $(x, y) \in [0, 1]^2$. The nontrivial diagonal fixed points form a one-parameter family parameterized by $1 \leq r$, corresponding to a one-dimensional branch of

equilibria. Now we could linearize $\mathbf{G}$ around $\mathbf{x}_1$ to see whether $\mathbf{x}_1$ is an isolated root. We do this by computing the Jacobian (for $x = y = t$ and $r_1 = r_2 = r$):

$$\begin{aligned}
\frac{\partial \mathbf{G}}{\partial \mathbf{x}} &= \frac{\partial \mathbf{F}}{\partial \mathbf{x}} - \frac{\partial \mathbf{x}}{\partial \mathbf{x}} && (3.25)\\
&= \frac{\partial \mathbf{F}}{\partial \mathbf{x}} - I && (3.26)\\
&= \begin{bmatrix} (1-\varepsilon)r(1-2t) & \varepsilon r(1-2t) \\ \varepsilon r(1-2t) & (1-\varepsilon)r(1-2t) \end{bmatrix} - I && (3.27)\\
&= r(1-2t)\begin{bmatrix} (1-\varepsilon) & \varepsilon \\ \varepsilon & (1-\varepsilon) \end{bmatrix} - I && (3.28)\\
&= (2-r)\begin{bmatrix} (1-\varepsilon) & \varepsilon \\ \varepsilon & (1-\varepsilon) \end{bmatrix} - I && (3.29)\\
\lambda^2 - \operatorname{tr}\left(\frac{\partial \mathbf{G}}{\partial \mathbf{x}}\right)\lambda + \det\left(\frac{\partial \mathbf{G}}{\partial \mathbf{x}}\right) &= 0 && (3.30)\\
\lambda_\pm &= \frac{1}{2}\operatorname{tr}\left(\frac{\partial \mathbf{G}}{\partial \mathbf{x}}\right) \pm \sqrt{\frac{1}{4}\operatorname{tr}\left(\frac{\partial \mathbf{G}}{\partial \mathbf{x}}\right)^2 - \det\left(\frac{\partial \mathbf{G}}{\partial \mathbf{x}}\right)} && (3.31)\\
&= (2-r)(1-\varepsilon) - 1 \pm \varepsilon(2-r) && (3.32)\\
\lambda_+ &= 1 - r && (3.33)\\
\lambda_- &= (2-r)(1-2\varepsilon) - 1 && (3.34)
\end{aligned}$$

Since $\det\left(\frac{\partial \mathbf{G}}{\partial \mathbf{x}}\right) = \lambda_+\lambda_-$, we know by the inverse function theorem that $\mathbf{x}_1$ is an isolated zero to $\mathbf{G}$ as long as neither $\lambda_+ = 0$ or $\lambda_- = 0$, otherwise the linearization becomes inconclusive and another branch, possibly off-diagonal can bifurcate from $\mathbf{x}_1$.

3.1.2. Antisymmetric fixed points

To see whether we have fixed points for the case $r_1 \neq r_2$ we can consider the uncoupled case $\varepsilon = 0$

$$\begin{bmatrix} r_1 & 0 \\ 0 & r_2 \end{bmatrix}\begin{bmatrix} x(1-x) \\ y(1-y) \end{bmatrix} - \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{3.35}$$

Then,

$$\mathbf{x}_2(r_1, r_2) = \left(1 - \frac{1}{r_1}, 1 - \frac{1}{r_2}\right) \tag{3.36}$$

which is off-diagonal and away from the singularity $\varepsilon = 1/2$ found in the determinant in Equation (3.9).

3.1.3. Perturbation theory & Antisymmetric fixed points

Let

$$r_1 = r_0 + \delta r_1^{(1)} + \mathcal{O}(\delta^2), \qquad r_2 = r_0 + \delta r_2^{(1)} + \mathcal{O}(\delta^2),$$

and seek a perturbative solutions of the form

$$x(\delta) = x_0 + \delta x_1 + \mathcal{O}(\delta^2), \qquad y(\delta) = y_0 + \delta y_1 + \mathcal{O}(\delta^2).$$

to the fixed point equation.

Then

$$\begin{aligned} x(1-x) &= \big(x_0 + \delta x_1 + \mathcal{O}(\delta^2)\big)\Big(1 - x_0 - \delta x_1 + \mathcal{O}(\delta^2)\Big) \\ &= x_0(1-x_0) + \delta x_1(1-2x_0) + \mathcal{O}(\delta^2), \end{aligned} \tag{3.37}$$

$$y(1-y) = y_0(1-y_0) + \delta y_1(1-2y_0) + \mathcal{O}(\delta^2). \tag{3.38}$$

Now consider the fixed-point equations

$$P(x,y) = (1-\varepsilon)r_1 x(1-x) + \varepsilon r_2 y(1-y) - x = 0, \tag{3.39}$$
$$Q(x,y) = (1-\varepsilon)r_2 y(1-y) + \varepsilon r_1 x(1-x) - y = 0. \tag{3.40}$$

Substituting the perturbation expansions gives

$$\begin{aligned} (1-\varepsilon)\Big(r_0 + \delta r_1^{(1)}\Big)\Big(x_0(1-x_0) + \delta x_1(1-2x_0)\Big) + \varepsilon\Big(r_0 + \delta r_2^{(1)}\Big)\Big(y_0(1-y_0) + \delta y_1(1-2y_0)\Big) \\ = x_0 + \delta x_1 + \mathcal{O}(\delta^2), \end{aligned}$$

$$\begin{aligned} (1-\varepsilon)\Big(r_0 + \delta r_2^{(1)}\Big)\Big(y_0(1-y_0) + \delta y_1(1-2y_0)\Big) + \varepsilon\Big(r_0 + \delta r_1^{(1)}\Big)\Big(x_0(1-x_0) + \delta x_1(1-2x_0)\Big) \\ = y_0 + \delta y_1 + \mathcal{O}(\delta^2). \end{aligned}$$

Collecting terms of order $\mathcal{O}(1)$ and $\mathcal{O}(\delta)$ yields the system

$$\begin{aligned} (1-\varepsilon)r_0\, x_0(1-x_0) + \varepsilon r_0\, y_0(1-y_0) &= x_0, \\ (1-\varepsilon)r_0\, y_0(1-y_0) + \varepsilon r_0\, x_0(1-x_0) &= y_0, \\ (1-\varepsilon)r_0\, x_1(1-2x_0) + \varepsilon r_0\, y_1(1-2y_0) + (1-\varepsilon)r_1^{(1)}x_0(1-x_0) + \varepsilon r_2^{(1)}y_0(1-y_0) &= x_1, \\ (1-\varepsilon)r_0\, y_1(1-2y_0) + \varepsilon r_0\, x_1(1-2x_0) + (1-\varepsilon)r_2^{(1)}y_0(1-y_0) + \varepsilon r_1^{(1)}x_0(1-x_0) &= y_1. \end{aligned}$$

For $\mathcal{O}(\delta)$ we have, $c_1 = (1-\varepsilon)r_0$, $c_2 = \varepsilon r_0$, $\sigma = r_1^{(1)} + r_2^{(1)}$, $t = 1 - 1/r_0$ and $\Delta = r_2^{(1)} - r_1^{(1)}$

$$\begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} c_1(1-2t) - 1 & c_2(1-2t) \\ c_2(1-2t) & c_1(1-2t) - 1 \end{bmatrix}\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = -u(t)\begin{bmatrix} r_1^{(1)} + \varepsilon\Delta \\ r_2^{(1)} - \varepsilon\Delta \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \tag{3.41}$$

$$\begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = VDV^{-1}\mathbf{x} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \tag{3.42}$$

$$\mathbf{x} = VD^{-1}V^{-1}\mathbf{b} = VD^{-1}V^T\mathbf{b} \tag{3.43}$$

$$\lambda_\pm = \alpha \pm \beta = (c_1 \pm c_2)(1-2t) - 1 \tag{3.44}$$
$$\lambda_+ = (c_1 + c_2)(1-2t) - 1 = r_0(1-2t) - 1 = (1-r_0) \tag{3.45}$$
$$\lambda_- = (c_1 - c_2)(1-2t) - 1 = (1-2\varepsilon)r_0(1-2t) - 1 = (1-2\varepsilon)(2-r_0) - 1 \tag{3.46}$$
$$\mathbf{e}_\pm = \frac{1}{\sqrt{2}}(\pm 1, 1) \tag{3.47}$$

This means that $a_\mathbf{e}_$ forces $\mathbf{x}$ away from the diagonal (smoothly).

$$V = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, D^{-1} = \begin{bmatrix} \frac{1}{\lambda_+} & 0 \\ 0 & \frac{1}{\lambda_-} \end{bmatrix} \tag{3.48}$$

So

$$\mathbf{x} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} \frac{1}{\lambda_+} & 0 \\ 0 & \frac{1}{\lambda_-} \end{bmatrix}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \tag{3.49}$$

$$x_1 = \frac{b_1 + b_2}{2\lambda_+} - \frac{b_2 - b_1}{2\lambda_-} \tag{3.50}$$

$$y_1 = \frac{b_1 + b_2}{2\lambda_+} + \frac{b_2 - b_1}{2\lambda_-} \tag{3.51}$$

$$b_1 + b_2 = -u(t)\sigma \tag{3.52}$$

$$b_2 - b_1 = -u(t)(1 - 2\varepsilon)\Delta \tag{3.53}$$

$$x_1 = -\frac{1}{2\lambda_+}u(t)\sigma + \frac{1}{2\lambda_-}u(t)(1 - 2\varepsilon)\Delta = u(t)\left(-\frac{1}{2\lambda_+}\sigma + \frac{1}{2\lambda_-}(1 - 2\varepsilon)\Delta\right) \tag{3.54}$$

$$y_1 = -\frac{1}{2\lambda_+}u(t)\sigma - \frac{1}{2\lambda_-}u(t)(1 - 2\varepsilon)\Delta = u(t)\left(-\frac{1}{2\lambda_+}\sigma - \frac{1}{2\lambda_-}(1 - 2\varepsilon)\Delta\right) \tag{3.55}$$

$$x_1 = -\frac{u(t)}{2}\left(\frac{\sigma}{\lambda_+} + \frac{(1 - 2\varepsilon)\Delta}{\lambda_-}\right) \tag{3.56}$$

$$y_1 = -\frac{u(t)}{2}\left(\frac{\sigma}{\lambda_+} - \frac{(1 - 2\varepsilon)\Delta}{\lambda_-}\right) \tag{3.57}$$

$$\mathbf{x}_1 = -\frac{u(t)\sigma}{2\lambda_+}\mathbf{e}_+ - \frac{u(t)(1 - 2\varepsilon)\Delta}{2\lambda_-}\mathbf{e}_- \tag{3.58}$$

Thus, the total perturbation of the diagonal fixed point is given by:

$$\mathbf{x}(\delta) = \mathbf{x}_0 + \delta\mathbf{x}_1 + \mathcal{O}(\delta^2) = t\mathbf{e}_+ + \delta\left(-\frac{u(t)\sigma}{2\lambda_+}\mathbf{e}_+ - \frac{u(t)(1 - 2\varepsilon)\Delta}{2\lambda_-}\mathbf{e}_-\right) + \mathcal{O}(\delta^2) \tag{3.59}$$

Which satisfies the fixed point equation up to first order as long as $\lambda_\pm \neq 0$ which are the same conditions the jacobian needs to satisfy to achieve invertibility on the diagonal fixed point.

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3.2. Problem 3: Phase Portraits

In general, a phase portrait displays the evolution of the state of a dynamical system by plotting subsequent states in the same diagram. For one-dimensional continuous systems the axes would usually be $\dot{x}$ and x. For more than one dimension or state variable one has to choose them to be well representable in two dimensions. In the system examined here we have an iterated map with two dimensions which are also the axes in the phase portrait plot.

If an attractive fixed point exists in the system, later iterations would be ever closer to that fixed point than previous iterations. A stable orbit of length n could not be distinguished from n fixed points being present and would be needed to be further examined by the other methods employed in this report. Chaotic conditions can be inferred when there is no discernible pattern in the phase portrait. Pattern could here refer to either convergence or regularity.

3.3. Bifurcation cascade theory

A bifurcation refers to a qualitative change in the behavior of a dynamical system as a parameter is varied. In the context of our system we are interested in how the number of stable orbits reacts to varying one of the r parameters while keeping other parameters constant. At low r values the system has only one stable orbit. At a certain r value this stable orbit will split in to two undergoing a period doubling bifurcation. A r is increased more these orbits will move further away from eachother and eventually undergo another period doubling. The step in r value between these bifurcations will keep getting smaller with each bifurcation. After reaching a certain r value the number of stable orbits will approach infinity and the system can be said to be chaotic. ======= »»»> a8aee4615a8ffd2669f39453c39fb93193ec01a2

4. Numerical method

In this section we will describe the numerical procedures used in our code and state the relevant parameters to reproduce our findings.

4.1. Finding fixed points in periodic orbits with Newtons Method

Newton's Method

To find the fixed points and periodic orbits we apply Newton's method. It is an iterative numerical procedure for solving nonlinear equations. In one dimension, it is used to solve equations of the form

$$g(x) = 0,$$

which perfectly represents the problem at hand.

Starting from an initial approximation x_0, Newton's method constructs a sequence of approximations according to

$$x_{n+1} = x_n - \frac{g(x_n)}{g'(x_n)}.$$

The method is based on a local linear approximation of the function. At each iteration, the function g is approximated by its tangent line at the current point x_n, and the zero of this tangent line is used as the next approximation. When the initial guess is sufficiently close to a true root and the derivative is nonzero, Newton's method converges.

In the present problem, the unknown is a two-dimensional vector, where

$$z = \begin{pmatrix} x \\ y \end{pmatrix}.$$

The function to be solved is therefore:

$$G_p(x, y) = \begin{pmatrix} G_{p,1}(x, y) \\ G_{p,2}(x, y) \end{pmatrix}.$$

Newton's method can then be applied through a system of two nonlinear equations,

$$\begin{aligned} G_{p,1}(x, y) &= 0, \\ G_{p,2}(x, y) &= 0. \end{aligned}$$

In this case, naturally, the derivative needs to be replaced by the Jacobian matrix of G_p:

$$J_p(x, y) = DG_p(x, y) = \begin{pmatrix} \frac{\partial G_{p,1}}{\partial x} & \frac{\partial G_{p,1}}{\partial y} \\ \frac{\partial G_{p,2}}{\partial x} & \frac{\partial G_{p,2}}{\partial y} \end{pmatrix}.$$

At each iteration, the Newton correction s_n is obtained by solving the linear system

$$J_p(x_n, y_n)s_n = -G_p(x_n, y_n).$$

The approximation is then updated according to

$$\begin{pmatrix} x_{n+1} \\ y_{n+1} \end{pmatrix} = \begin{pmatrix} x_n \\ y_n \end{pmatrix} + s_n.$$

In the implemented code, the Jacobian matrix is computed symbolically, and at each Newton iteration the corresponding linear system is solved numerically. This produces a correction vector that moves the current approximation closer to a root of G_p. In the present implementation, the damping parameter is set equal to 1, corresponding to the standard Newton method.

Newton's method is a local method, meaning that its convergence depends strongly on the choice of initial condition. Different initial guesses may converge to different roots, fail to converge, or lead to points outside the meaningful region. For this reason, the program applies Newton's method to a grid of initial points in the unit square.

After the Newton iteration terminates, the resulting point must be verified. This is necessary because a small Newton step does not always guarantee that the point is an actual root of G_p. The program therefore evaluates the residual

$$\|G_p(x, y)\|.$$

A point is accepted only if this residual is below a defined tolerance. This ensures that the computed point satisfies

$$F^p(x, y) \approx (x, y).$$

In addition, the program only accepts solutions within the meaningful domain, the unit square values,

$$0 \leq x \leq 1, \qquad 0 \leq y \leq 1.$$

Finally, the minimal period of the point is checked. This step is essential because the solutions may include points of smaller period. For example, fixed points are also solutions of the period-2 equation. Therefore, the code tests the smallest integer $k \leq p$ for which

$$F^k(x, y) \approx (x, y).$$

Only points whose minimal period is exactly p are retained as actual period-p points.

4.2. Creating Phase Portraits

As explained in the theory part, to create a phase portrait the map needs to be iterated. To try to better capture the full phase space multiple random points are sampled from the state space as starting points. For each starting point the map is first iterated for 30 steps as a warm-up to have a cleaner plot. After this, the map is further iterated for 200 more iterations and all points are plotted into the x-y plane. To improve the ability to recognize patterns, a colour gradient is used for each collection of points. This colour then serves as a hint to see how the location evolves over multiple iterations and makes, for example, the convergence towards a fixed point or the irregularity of the trajectory better visible. The figures from the results section for this subtask can be recreated by running the Python file `phasePortrait.py` handed in with the report as a script.

4.3. Bifurcation Cascade

For implementation of the map we use the function `coupled_map` from `common.py`. This function calculates the next generation of x and y values from the given x and y values, r_1, r_2 and ϵ. As we need more than one time step, we created the function `calc_timeline`, which returns the x and y values for a certain r_2 value. This function also has the option to choose different numbers of steps and discard a certain number of timesteps from the beginning of the time series.

Additionally, we set our constants to the values given in the task: $r_1 = 3.1$ and $\varepsilon = 0.3$. For r_2 we created a linspace from 2.8 to 3.97 with 800 values. The x and y values each iteration starts from are set to $x_0 = 0.2$ and $y_0 = 0.6$. We decided to use 400 iterations and discard the first 150. We set these values at the beginning of our work and were satisfied with the results.

For each value in the r_2 linspace we then calculate the time series and save the x values. We then use `plt.plot` to plot all these values and then move to the next r_2 value. Lastly, we added labels and the title and saved the figure.

To obtain the bifurcation diagram (as in Figure 9) you need to run `bifurcationCascade.py`. The figure will be displayed and saved in the folder 'figures'. Thus, there must be a 'figures' folder where you run the code.

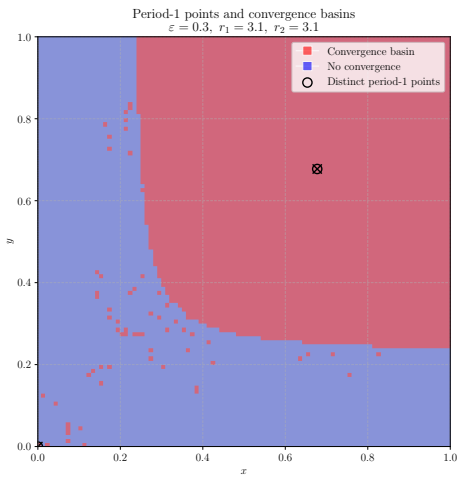
5. Results

5.1. p-Orbits

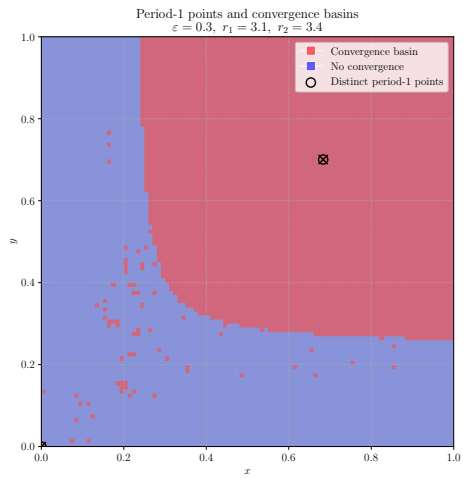
Figure 1 displays the period-1 points found by solving $F(x, y) = (x, y)$ for $\varepsilon = 0.3$, $r_1 = 3.1$, and different values of r_2. For the symmetric case, $r_1 = r_2 = 3.1$, which goes accordingly with the theoretical prediction, where

$$x = y = 1 - \frac{1}{3.1}.$$

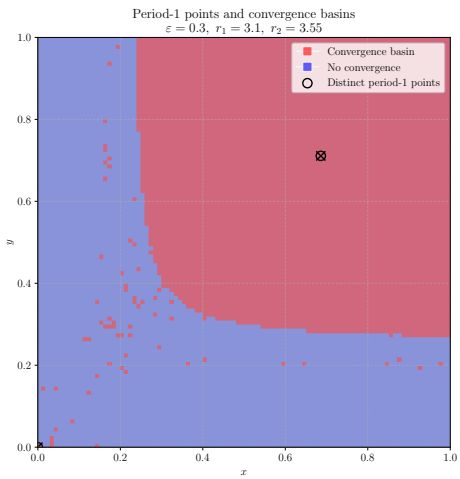
One can also verify that, when one increases r_2 and r_1 is kept fixed, the fixed point moves away from the diagonal $x = y$. In terms of physical interpretation, it is to be expected since now the growth parameter differs between both populations. The second population grows more rapidly, which can be observed by the stronger increase in the y-coordinate. the



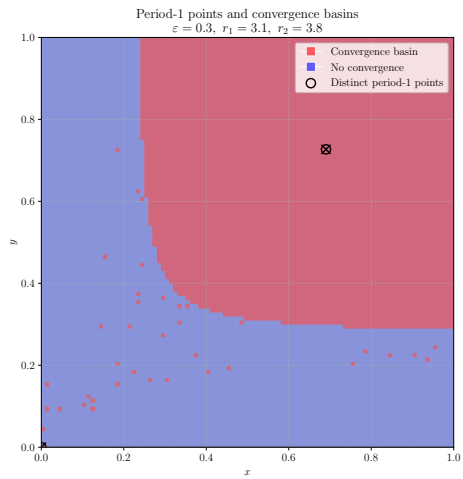
(a) `[0.67741935 0.67741935]`



(b) `[0.68345666 0.70051177]`



(c) `[0.68615866 0.71094441]`



(d) `[0.69025989 0.72689488]`

Figure 1. p-orbit behaviour for first parameter sweep

The existence of these fixes points does not, however, describe the long-term behavior of the system. For that, one has to further study their stability, which can be achieve through the following

period-p points and the respective bifurcation diagram.

5.1.1. 2-orbits

Figure 2 shows the points that satisfy

$$F^2(x, y) = (x, y)$$

with minimal period 2. This minimal condition, as observed previously, is fundamental for the study and identification of genuine period-p points. Although they satisfy $F^2(x, y) = (x, y)$, they do not count as genuine period-2 orbits. For the symmetric case $r_1 = r_2 = 3.1$, two period points are found,

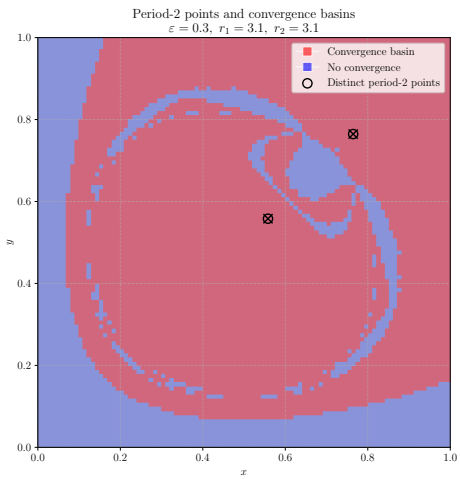
$$(0.76456652, 0.76456652) \quad \text{and} \quad (0.55801413, 0.55801413),$$

explains why the phase portrait for the same parameters shows two attracting locations rather than convergence to the period-1 fixed point.

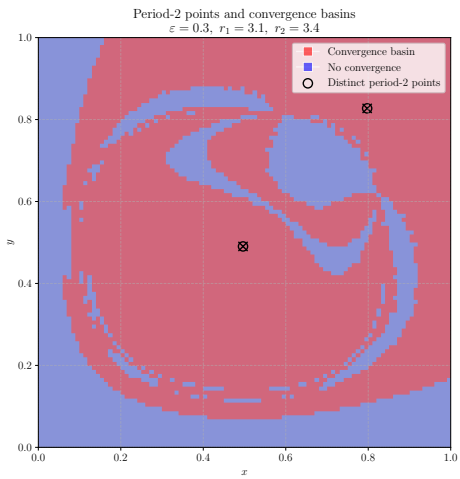
For the assymetric case, $r_2 > r_1$, two fixed points were found and observed to move away from the diagonal, where the displacement increases with r_2. One can easily observe this comparing $r_2 = 3.55$ and $r_2 = 3.8$, where the separation between the two points becomes larger. This is consistent with the bifurcation diagram, where the single branch splits into two and four branches, respectively.

5.1.2. 3-orbits

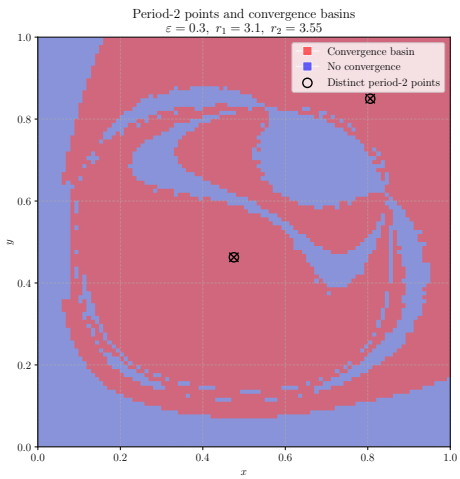
Following the project instructions, unlike initially expected, no genuine period-3 orbits were detected. This is a very particular case where it was observed that every fixed point satisfies $F^3(x, y) = (x, y)$, and none is minimal, so those cannot be counted as period-3 orbits.



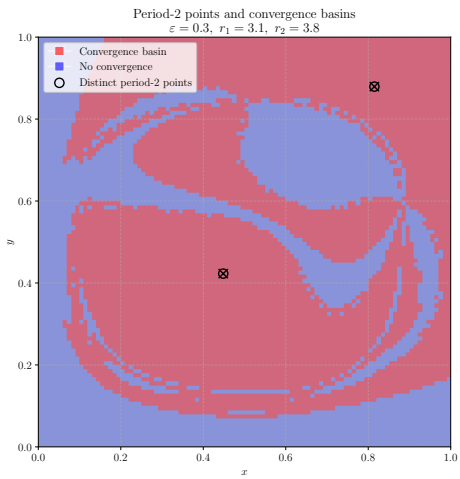
(a) `[0.76456652 0.76456652]`, `[0.55801413 0.55801413]`



(b) `[0.79737624 0.82726612]`, `[0.49635601 0.49035224]`



(c) `[0.80595371 0.84973562]`, `[0.47535588 0.46274208]`



(d) `[0.8149268 0.87917722]`, `[0.4483779 0.42282117]`

Figure 2. p-orbit behaviour for second parameter sweep

5.1.3. 4-orbits

Following the given suggestion, annother period was tested, period-4. Which was found to describe the behavior of the system as desired. Figure 3 shows a genuine period-4 orbit for $(r_1, r_2, \varepsilon) = (3.1, 3.8, 0.3)$. This now goes along with the bifurcation diagram that exhibits four points at approximately the same value of r_2.

ow a clearer description of the systems is achieved, the system cycles through four distinct states before repeating. This is a typical non-linear systems where the fixed pints change with the parameters exhibiting double-periodic dynamics.

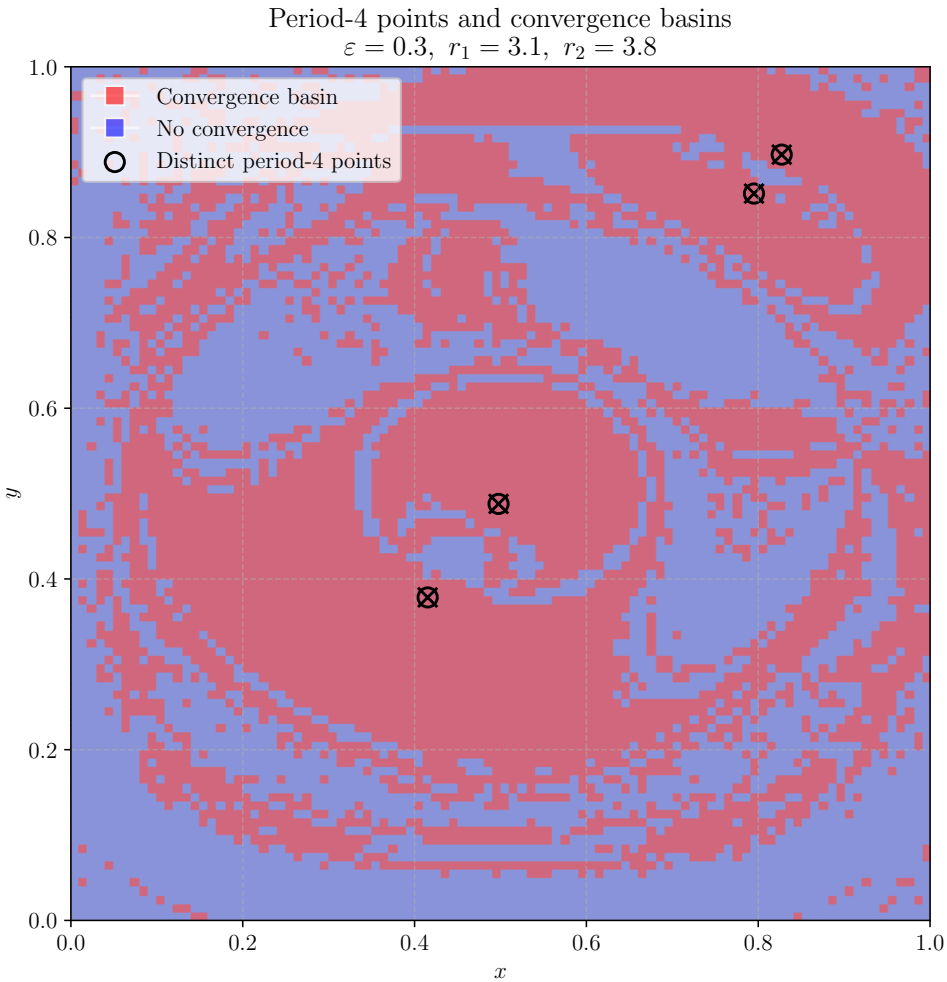


Figure 3. `[0.82732296 0.89710766], [0.49775917 0.48792763], [0.41523379 0.37839216], [0.79504902 0.85148034]`

5.2. Phase Portraits

This part of the report shows a total of five different figures with four different regimes, three of the parameter sets were taken from the suggestions in the lab instructions while two were to further illuminate the two-dimensional behaviour of the system at different points in the bifurcation diagram.

For the phase portrait shown in Figure 4, one can see clear convergence towards a single fixed point. Due to the removal of a transient period before plotting the evolution, the plot extents are quite small. Convergence can be seen because the points away from the yellow dot indicating the final convergence of the last evolution are quite dark and therefore need to come from the beginning of a cycle. Towards the dot there is also a faint brightening of the colour showing that the later iterations end up closer to the convergence point.

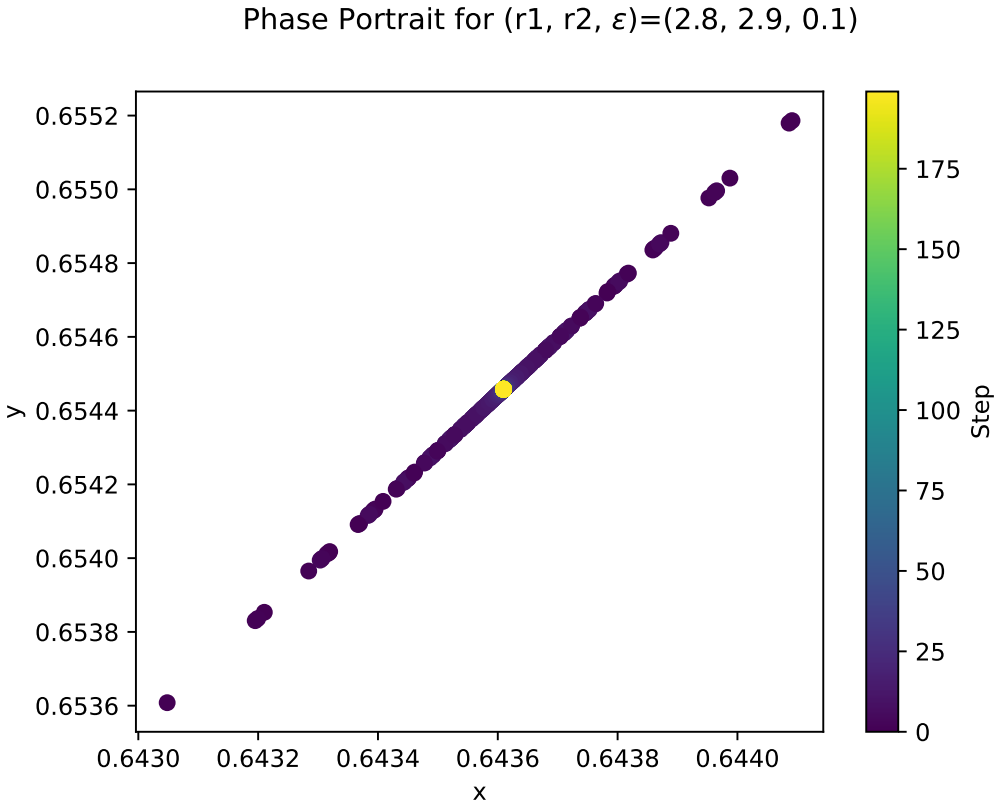


Figure 4. Phase Portrait for $(r_1, r_2, \epsilon) = (2.8, 2.9, 0.1)$

Figure 5 shows the first appearance of a bifurcation in the phase portrait. This bifurcation means that now two different attractive points exist instead of only one. As stated in the methods section, a phase portrait like this does not allow to make the distinction between two attractive, fixed points and a period-2 orbit. As expected for the symmetric case $r_1 = r_2$, both points lie on the diagonal with $x = y$.

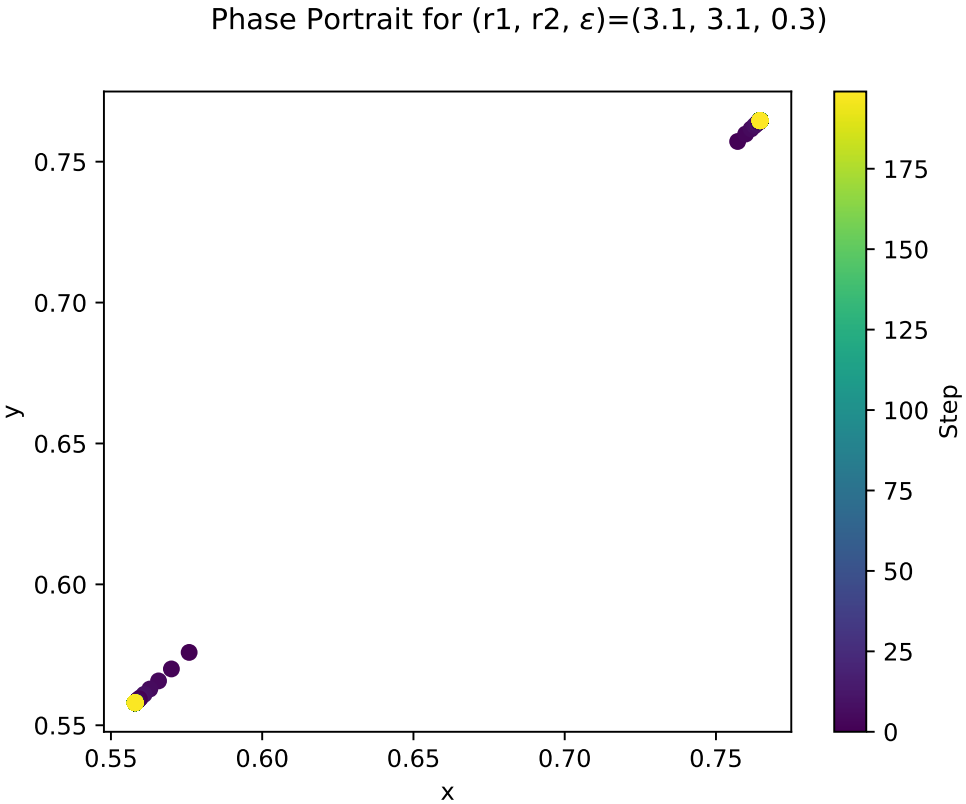


Figure 5. Phase Portrait for $(r_1, r_2, \epsilon) = (3.1, 3.1, 0.3)$

The impact of varying only one parameter can be seen in Figure 6. When raising r_2 slightly, the bifurcation into two attractive basins remains, while the locations shift. With this configuration the two points are farther apart and slightly away from the aforementioned diagonal. Due to the medium strong coupling the displacement is only barely recognizable in the plot.

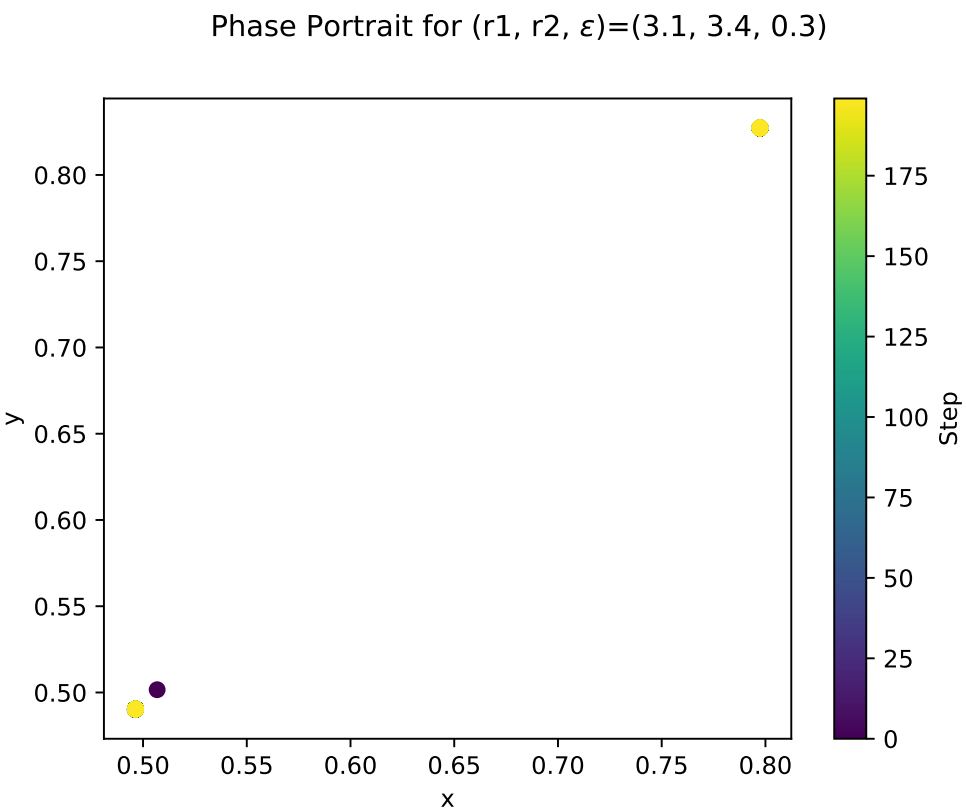


Figure 6. Phase Portrait for $(r_1, r_2, \epsilon) = (3.1, 3.4, 0.3)$

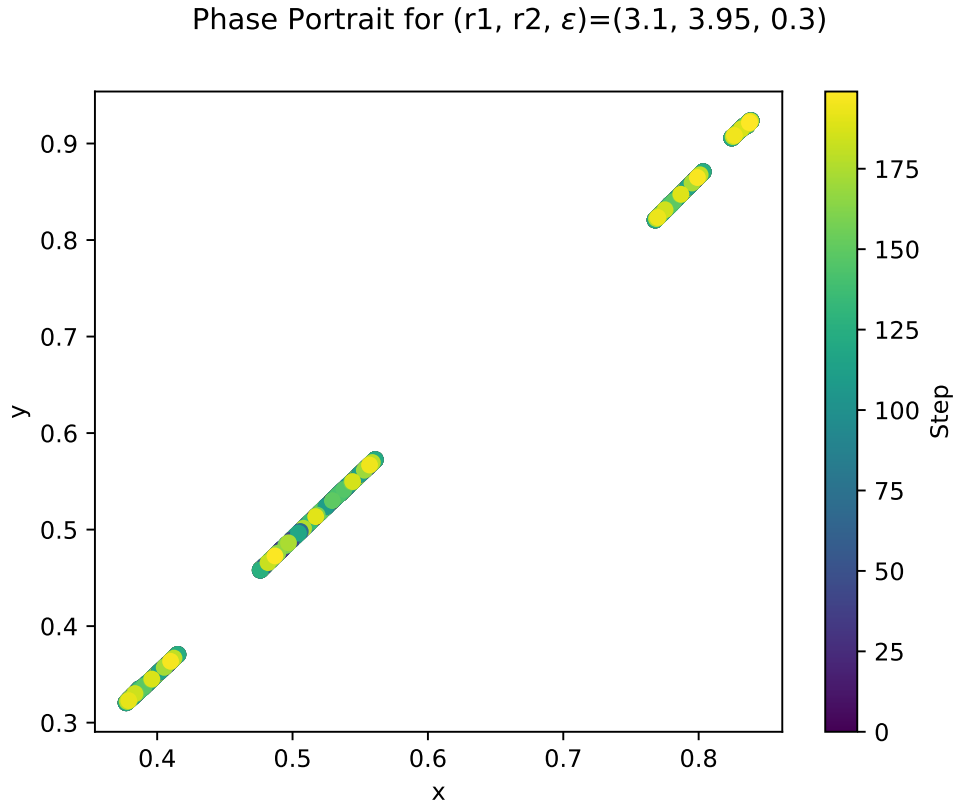


Figure 7. Phase Portrait for $(r_1, r_2, \epsilon) = (3.1, 3.95, 0.2)$

When further increasing r_2, we reach the chaotic region of r for the simple logistic map. The impact on the investigated coupled map is shown in Figure 7. Here we can identify four distinct region in the phase portrait, all of which do not have a single point of convergence inside them but a certain spread and seem to be aligned along a common line. The sizes of the regions vary significantly, the smallest is in the top right corner at around $(x, y) = (0.84, 0.92)$ and the largest region as at approximately $(x, y) = (0.53, 0.54)$ with a spread of around 0.05 in both coordinates, slightly more so in the y-direction than the x-direction.

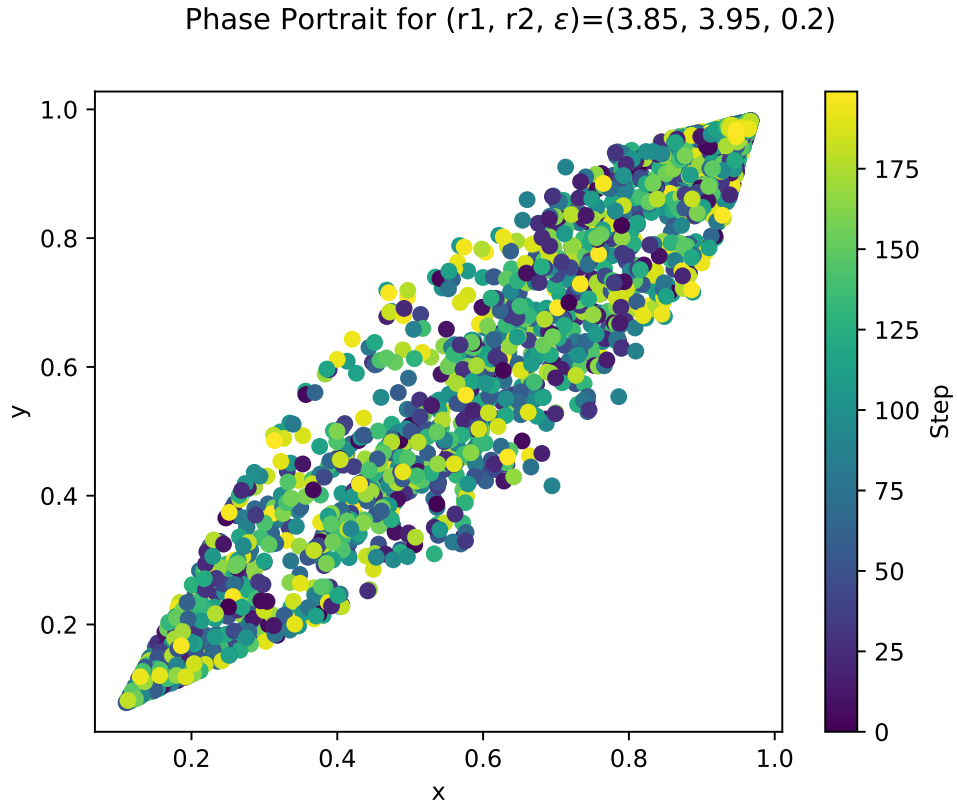


Figure 8. Phase Portrait for $(r_1, r_2, \epsilon) = (3.85, 3.95, 0.2)$

The best picture of a chaotic behaviour is Figure 8. Here, there is no discernible correlation in the distribution of points between the location in the plane and the temporal position in the evolution. All the points together form a kind of almond shape which is wider in the centre and almost a point at both ends of the shape. The shape is oriented almost along the diagonal but not quite because the scaling parameters r_1 and r_2 are once again chosen to be slightly different, however, both in the chaotic regime.

5.3. Bifurcation Cascade

Figure 9 shows the bifurcation cascade of the coupled logistic map. At the lowest r_2 values, from 2.8 to around 2.87, we can see one branch at an (almost) continuous value. This denotes one fixed point for these r_2 values. From $r_2 \approx 2.87$ the branch starts splitting into two and these branches separate from each other. At $r_2 \approx 3.7$ both branches split again so that we have four different branches. Each branch splits again at $r_2 \approx 3.9$. At slightly higher r_2 values the branches start splitting faster so that we end up with a point cloud that grows larger with higher r_2 values. Here, we will not observe fixed points.

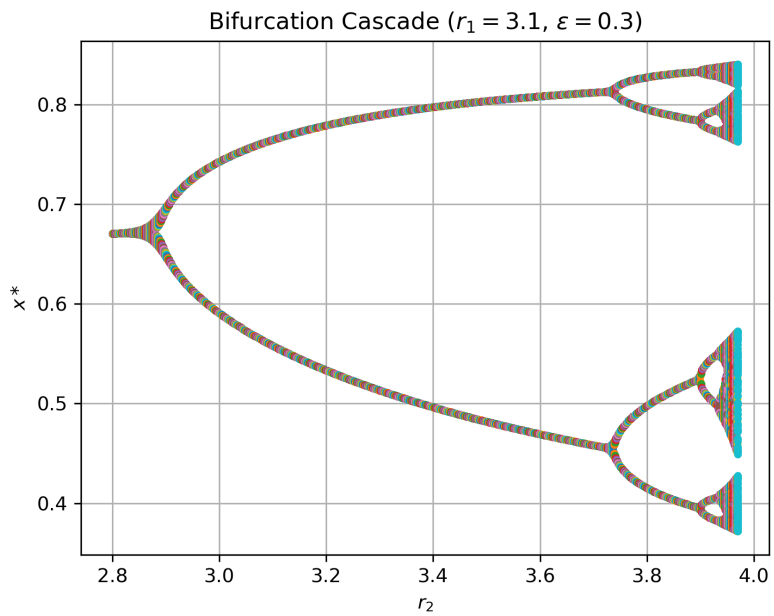


Figure 9. Bifurcation cascade for $r_1 = 3.1$ and $\epsilon = 0.3$

6. Interpretation

Now, we would like to combine the results of the different tasks and see if they give a coherent picture of the logistical map. For all three different tasks there is data on $r_1 = 3.1$ available and thus this case will be used for comparison.

For the symmetrical case ($r_1 = r_2 = r$) we expect a fixed point at $x = y = 1 - \frac{1}{r}$, which in this case corresponds to $x = y = 1 - \frac{1}{3.1} = 0.67741935$. Looking at the period-1 points in Figure 1 we can see that the $r_1 = r_2 = 3.1$ period-1 fixed point is exactly at the expected $x = y = 0.67741935$. This matches our theoretical fixed point very well. However, we do not see this fixed point in the corresponding phase portrait (Figure 5) or the bifurcation cascade (Figure 9). The reason for this is that the fixed point does not seem to be stable or attractive. No starting values of x and y converge to this fixed point. Instead, both the phase portrait and the bifurcation cascade show a convergence towards two different points ($x = y = 0.76456652$ and $x = y = 0.55801413$). For the bifurcation cascade this is represented in the two strings, while for the phase portrait we can see two yellow data points. As we can see in Figure 2, these two points are the period-2 points for these system parameters. The period-2 points are stable and attractive. Higher order orbits were not observed for these parameters.

Next, we can look at the asymmetrical case where $r_1 = 3.1 \neq r_2$. There are several interesting values of r_2 that show different behaviour. At $r_2 = 3.4$ there are still two branches in the bifurcation cascade, which hints at stable and attractive period-2 points. In this case, however, the points are not symmetrical. As we can tell from Figure 2 the coordinates of the fixed points are [0.79737624 0.82726612] and [0.49635601 0.49035224]. As r_2 is now larger than r_1, the y coordinates of the period-2 points moved faster away from the period-2 points for $r_1 = r_2 = 3.1$ than the x coordinates. These stable period-2 points and their asymmetric nature can also be seen in the phase portrait in Figure 6.

At $r_2 = 3.8$ the bifurcation cascade already exhibits four different branches. This is in agreement with the period-4 point that exists at this r_2 value (Figure 3). Again, this seems to be the stable and attractive fixed point instead of the period-2 or period-1 fixed points.

At $r_2 = 3.95$ the bifurcation cascade shows a point cloud. Similarly, the phase portrait (Figure 7) exhibits multiple different and well distributed points at higher step numbers. Interestingly, there seems to be a linear relation between the x and y value. Looking at the phase portrait for $r_1 = 3.85$, $r_2 = 3.95$ this relation is not given at higher r_1 values. Instead, for each x value there is a cloud of y values.

All in all, the four tasks show a coherent picture. In the symmetrical case there is a fixed point at $x = y = 1 - \frac{1}{3.1} = 0.67741935$, which is, however, unstable and thus not visible in the bifurcation cascade or the phase portrait. Instead, the logistical map converges towards period-2 points. At higher r_2 we exhibit stable and attractive period-4 points and, at even higher values, a point cloud. In all cases, the results of the p-Orbits, the bifurcation cascade and the phase portraits match.

7. Conclusion

In this project, we performed an in-depth study of a two-dimensional discrete-time dynamical system given by two logistic maps. The aim was to investigate how the long-term behaviour of the system changes for different parameter values, using fixed points, periodic points, phase portraits and a bifurcation diagram.

We have presented different developments of the system depending on the initial parameters. For smaller parameter values, trajectories may converge to a single attracting fixed point. The theoretical fixed points were calculated and match the later obtained results with the exception of the symmetric case $r_1 = r_2 = 3.1$, where $x = y = 1 - \frac{1}{3.1} = 0.67741935$ theoretically exists. However, for these values of r_1 and r_2 this value is not observed in the numerical simulations as it is not attractive. Instead, the system converges to a stable period-2 orbit, from what we can conclude that the existence of a fixed point alone is not enough to describe the observed long-term behaviour. It is therefore clear that the fixed point's stability is essential to describe the behaviour of the system.

The symmetry between the two populations is broken when increasing r_2 while keeping $r_1 = 3.1$. In this case, the system still exhibits a period-2 behaviour which turns into an asymmetric orbit. For the tested parameter values, no genuine period-3 orbit was detected after checking the minimal period of the numerical solutions. For larger values of r such as $r_2 = 3.8$, the bifurcation cascade and the computation of periodic points both indicate a stable period-4 orbit. With even greater values, such as $r_2 = 3.95$, the system no longer converges to a small number of points, but instead produces a point cloud, suggesting more complex and chaotic dynamics.

Overall, the different numerical approaches give a consistent picture. The numerical calculations, phase portraits and bifurcation diagram all show a transition from simple attracting behaviour to period-doubling and then to increasingly complex dynamics. This leads to the conclusion that deterministic systems can become highly complex, especially when two logistic maps are coupled, generating a behaviour that appears irregular or aperiodic despite being produced by deterministic rules.

8. Use of LLM tools

For Task 2, ChatGPT was used as a debugging and code-refinement tool. The core programming work, including the implementation of Newton's method, the search for period-2 and period-3 points, and the production of the corresponding plots, was carried out by us. When comparing the numerical results with those obtained in other parts of the assignment, we noticed that the code was returning more periodic points than expected. By uploading the code and using prompts such as "Check the Jacobian evaluation, update step, stopping criterion, and handling of failed iterations" and "Numerically identical solutions are counted multiple times, will a tolerance-based comparison work better than exact floating-point equality?", we identified and corrected several numerical and logical issues in the implementation. This led to the addition of a tolerance-based duplicate filter, a minimal-period verification step, a direct numerical iteration routine, and safer convergence handling within Newton's method. No LLM was used for Tasks 3 and 4.

Files submitted

List the files included in the submission.

References

[1] Jordi-Lluís Figueras. Lecture slides modelling complex systems, 2026.